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Praesignis AWS re/Start Special Projects

Project 1 - Building a Static Website for a Restaurant and Presenting AWS Migration Benefits

1. Project Overview

You are tasked with creating a static website for a real-world restaurant (you can choose an existing local restaurant or a small business). The restaurant is experiencing operational challenges due to a high volume of customer bookings and orders, which are getting mixed up, and the owner is struggling to find a solution. The owner wants to use a simple static website to streamline customer interactions and improve order and booking management. The website will be hosted using AWS services, leveraging the current lab setup for the static website. The project will be enhanced by adding a sign-in page using AWS Cognito and a back end with a database to manage customer data, bookings, and orders. Additionally, you will prepare a presentation to showcase the benefits of migrating the restaurant's digital presence to AWS, targeting a panel of your choice (e.g., the restaurant owner or a mock business panel).

2. Project Deliverables

2.1. ***Static Website:*** Build a fully functional static website for the restaurant, hosted on AWS with Cognito sign-in, a database back end for managing bookings and orders, and features like a booking form and order submission page.

2.2. ***PowerPoint Presentation:*** Create a presentation (slides and notes) covering the restaurant's overview, challenges, AWS services, cost analysis, and benefits of migrating to AWS.

3. Project Requirements

3.1. Static Website Development:

3.1.1. Build a ***simple static website*** for the restaurant using the current lab setup.

3.1.2. Add a ***sign-in page*** using AWS Cognito for user authentication, allowing customers to log in and manage their bookings or orders.

3.1.3. Implement a ***back end*** with a database (e.g., Amazon RDS or DynamoDB) to store customer data, bookings, and order details.

3.1.4. Optionally, ***store website images*** (e.g., menu photos or restaurant ambiance) in the database if feasible; otherwise, use Amazon S3 for image hosting.

3.1.5. Ensure the website is ***user-friendly***, with a ***clean design*** to help customers easily make bookings or place orders. Include ***features*** like a booking form, an order submission form, and a confirmation page.

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3.1.6. Use ***minimal colours*** to keep text readable, but incorporate images, videos, or animations to enhance the website's design (e.g., photos of the restaurant, a short video of the dining experience, or an animated menu display).

3.2. AWS Services Presentation

3.2.1. Company Overview

The restaurant (e.g., "Tasty Bistro," a local eatery) is a popular dining spot experiencing a surge in customers. They receive a high volume of bookings and orders daily, both in-person and via phone or email. However, their current manual system (e.g., paper-based or basic spreadsheet tracking) leads to frequent mix-ups—orders get confused, bookings are double-booked, and the owner struggles to track customer preferences or confirmations. The owner wants to implement a simple static website to allow customers to make bookings and place orders online, reducing errors and improving efficiency.

3.2.2. Restaurant Challenges

- The restaurant has not migrated to the cloud and relies on manual or on-premises systems (e.g., paper records or a local computer) to manage bookings and orders.
- This leads to operational inefficiencies: order mix-ups (e.g., wrong dishes delivered), double bookings, and delays in confirming customer requests.
- The owner lacks a centralised system to track customer data, making it hard to identify repeat customers or manage special requests.
- Scaling the current system to handle more customers is costly and time-consuming (e.g., hiring more staff to manage orders or upgrading on-premises hardware).

3.2.3. AWS Services: Recommend AWS services to address the restaurant's needs:

- Use Amazon S3 to host the static website and store images (e.g., menu photos).
- Use AWS Cognito for user authentication, allowing customers to sign in and access their booking/order history.
- Use Amazon RDS or DynamoDB for the database to store customer profiles, booking details, and order information.
- Optionally, include AWS Lambda for serverless functions (e.g., to send automated confirmation emails after a booking or order is placed) and Amazon SNS for sending notifications to the restaurant staff about new orders.

3.2.4. Benefits: Explain why the restaurant should choose AWS.

3.2.5. Presentation Guidelines

- Conciseness of presentation; avoid overloading slides with information (use notes on the side for additional details).
- Use minimal colours to ensure text visibility, but include images, videos, or animations to make the presentation engaging (e.g., screenshots of the website’s booking form, a demo video of placing an order, or AWS service diagrams).